

THE MODERN ANIMAL NUTRITION LIBRARY

Nutritional Diseases in Farm Animals

Scientific Principles, Clinical Recognition, Diagnosis, Management
and Prevention

From nutrient imbalance to metabolic disturbance, physiological dysfunction, clinical signs and production loss, this volume follows one line of reasoning across cattle, dairy cows, poultry, fish, sheep and goats: understand the mechanism, recognize the disease early, correct the ration with evidence, and prevent recurrence by design.

Prof. Mousa M. A.

Professor of Nutrition and Clinical Nutrition
Faculty of Veterinary Medicine · Sohag University
Sohag, Egypt

CATTLE & DAIRY

POULTRY

FISH

SHEEP & GOATS

FIRST EDITION · 2026

VETERINARY NUTRITION SERIES

The Modern Animal Nutrition Library

NUTRITIONAL DISEASES IN FARM ANIMALS

Scientific Principles, Clinical Recognition, Diagnosis, Management and Prevention

Prof. Mousa M. A.

Professor of Nutrition and Clinical Nutrition
Faculty of Veterinary Medicine · Sohag University

First edition, 2026.

© 2026 Prof. Mousa M. A. All rights reserved.

No part of this publication may be reproduced, stored in a retrieval system,
or transmitted in any form or by any means, electronic, mechanical,
photocopying or otherwise, without the prior written permission of the author.

The author has taken reasonable care to ensure the accuracy of the information
presented. Dose rates, feed compositions and management recommendations must
always be adapted to local conditions, feed analysis results and professional
judgment. The educational case studies in this book are hypothetical.

Typeset in FreeSerif.

Sohag, Egypt.

About the Author

Prof. Mousa M. A. is Professor of Nutrition and Clinical Nutrition at the Faculty of Veterinary Medicine, Sohag University, Egypt. His teaching and professional work center on the nutrition of farm animals, the clinical recognition of nutrient imbalances, and the practical management of feeding programs under field conditions.

He is the author of The Modern Animal Nutrition Library, a series of digital reference books written for veterinary students, practicing veterinarians, animal scientists, nutritionists and livestock producers. The present volume grew out of a simple observation repeated in lecture halls, clinics and farm visits alike: most nutritional diseases are recognized late, although almost all of them can be prevented early.

Preface

Every farm has a silent employee whose work nobody schedules and whose mistakes nobody notices until the payroll shrinks. That employee is the ration. When it is balanced, the animals convert it into milk, meat, eggs or growth quietly and efficiently. When it is wrong, in a way that is sometimes too small to see on paper, the conversion stalls, resistance falls, and disease follows with a delay of days, weeks or an entire gestation. By the time a farmer points at a lame cow, a thin ewe or a tank of pale fry, the nutritional problem has usually been operating for a long time.

This book was written to shorten that delay. It follows one line of reasoning from the first ingredient to the last clinical sign: a nutrient becomes unbalanced; metabolism is disturbed; a physiological system loses control; tissue damage accumulates; signs appear; production is lost. Every disease in every chapter is explained along that same pathway, because clinicians who understand why a sign develops will also know which analysis to request, which other diseases look similar, and which correction will hold.

The book is organized by species, from cattle and dairy cows through poultry and fish to sheep and goats, and closes with the comparisons that tie the species together: a cross-species chapter, a diagnostic atlas arranged by clinical presentation, decision trees, hypothetical teaching cases and practical appendices designed to be photocopied and used in the farm office. Two convictions run through all of it. First, prevention is a formulation and management discipline, not an emergency measure. Second, no ration correction should ever be prescribed without knowing what the animals actually eat, what the feeds actually contain and what the laboratory actually says.

All cases are hypothetical teaching constructs, clearly labeled as such, and all dose questions are deliberately answered with principles plus ranges, because a book that hands out universal numbers would misserve the reader the moment conditions change. The reader should finish this volume with something more durable than a list of deficiency signs: a habit of thinking from nutrient to mechanism to sign, and back again.

Prof. Mousa M. A.
Sohag University, 2026

How This Book Is Organized

The seventeen chapters are grouped into six parts. Part I builds the vocabulary: what a nutritional disease is, how nutrients relate to organs and signs, and which risk factors decide whether a margin of error becomes a clinical case. Parts II through V cover the species in depth: cattle and dairy cows, poultry, fish, and the small ruminants. Part VI closes the circle with a cross-species comparison and the practical tools: decision trees, a diagnostic atlas arranged by clinical presentation, teaching cases and appendices for daily farm use.

Diseases are presented in a fixed order so that the reader always knows where to look: definition, importance, etiology, the normal role of the nutrient, pathogenesis, risk factors, clinical signs, pathological findings, laboratory findings, diagnosis, differential diagnosis, corrective nutrition, prevention and production consequences, and a short list of take-home messages. Major disorders receive the full treatment; minor ones are grouped in tables so that the reference stays readable.

Three kinds of boxed notes interrupt the text deliberately. Clinical Pearls carry the kind of judgment that usually comes only with years in the field. Nutritional Warnings mark the places where a well-meant correction, especially a supplement poured from a sack without analysis, makes the animal worse. Key take-home messages end each major disease so that a revision before an examination or a farm visit takes minutes, not hours.

The appendices are working documents: deficiency and toxicity summaries, antagonism tables, checklists for feed quality, farm audits and laboratory investigations, an investigation form, and conversion tables. The glossary at the end of the book defines the terms used throughout, and the reference list names the standards on which the scientific content rests.

Contents

Chapter 1 • Understanding Nutritional Diseases	1
What a Nutritional Disease Is	3
Why Nutritional Disease Is Recognized Late	4
The Economic and Welfare Weight	4
The Cascade from Imbalance to Prevention	5
A Working Vocabulary of Nutritional Disturbance	7
Nutrient Antagonism and Interaction	9
Prevention as the Destination	10
Chapter 2 • Nutrient-Disease Relationships	12
The Master Pathway from Nutrient to Sign	14
Energy	14
Protein and Amino Acids	15
Carbohydrates and Fiber	16
Lipids and Fatty Acids	16
Macrominerals	17
Trace Minerals	18
Vitamins	20
Water and Electrolytes	21
No Nutrient Acts Alone	22
Chapter 3 • Risk Factors for Nutritional Disease	24
Risk as the Meeting of Requirement and Supply	26
Animal Factors	26
Feed Factors	28
Environmental and Management Factors	29
Reading a Farm Through Its Risk Factors	32
Chapter 4 • Nutritional Diseases of Cattle	33
Energy Balance and Its Disorders	35
Protein and Rumen Nitrogen	37
Macromineral Disorders	38
Trace Mineral Disorders	49
Vitamin Disorders	58
Chapter 5 • Nutritional Diseases of the Dairy Cow	63
The heifer: growth and rearing	65

Pregnancy and fetal programming	66
The dry period	66
The transition period	67
Early lactation	69
Peak milk: nutrient partitioning	77
High DMI period: rumen health under load	77
Late lactation and body condition recovery	78
Chapter 6 • Rumen-Related Nutritional Disorders	80
The rumen as a fermentation system	82
Subacute ruminal acidosis (SARA)	83
Acute ruminal acidosis (grain overload)	85
Feed sorting and ration inconsistency	88
Rumen protein imbalance	89
Frothy bloat of grazing cattle	90
Reading the rumen: a daily discipline	90
Chapter 7 • Nutritional Diseases of Poultry	92
Energy disorders	95
Protein and amino acid disorders	96
Mineral disorders	98
Vitamin deficiencies	108
Chapter 8 • Nutritional Diseases of Laying Hens	116
Calcium metabolism of the laying hen	118
Shell quality disorders	119
Phosphorus and vitamin D in shell integrity	124
Yolk and albumen abnormalities	124
Production and egg mass decline	125
Chapter 9 • Nutrition and Chick Quality	127
Breeder nutrition and maternal transfer	130
Embryonic development and hatchability	131
Chick quality as a nutritional mirror	132
Early post-hatch nutrition	133
Chapter 10 • Nutritional Diseases of Fish	136
Why fish nutrition is different	138
Protein and amino acid disorders	138
Lipid disorders	140

Carbohydrate intolerance	141
Vitamin deficiencies	142
Mineral disorders	147
Chapter 11 • Feed Quality and Nutritional Disease in Aquaculture	149
Feed oxidation and rancidity	151
Mycotoxins in aquafeeds	152
Pellet quality, water stability and leaching	153
Feed contamination and adulteration	154
Storage and farm-level feed management	155
Chapter 12 • Nutritional Diseases of Sheep	158
Energy Disorders	160
Mineral Disorders	163
Copper: the Double-Edged Trace Element	165
Other Trace Elements	168
Rumen and Urinary Disorders	171
Nutrition of Reproduction and Lamb Growth	173
Chapter 13 • Nutritional Diseases of Goats	175
Species-Specific Nutrient Utilization	177
Pregnancy and Lactation Nutrition	177
Metabolic Disorders	178
Trace Mineral Disorders	180
Urinary Calculi	184
Kid Growth Problems	186
Chapter 14 • Cross-Species Comparison: One Nutrition, Six Animals	187
Energy Metabolism Across Species	189
Protein and Amino Acids	191
Fiber: Asset or Liability	191
Calcium, Phosphorus and the Skeleton	192
Selenium, Copper and Vitamin C: Three Great Contrasts	192
Water: the Forgotten Nutrient	193
Signature Diseases as a Mirror of Physiology	194
Chapter 15 • Clinical Decision Trees	196
The animal shows poor growth	198
Milk production falls below target	200
Eggshell quality deteriorates	202

Poor growth in a fish population	203
Neurological signs in the herd or flock	205
Chapter 16 • The Diagnostic Atlas by Clinical Presentation	208
Poor growth	210
Reduced feed intake	211
Reduced milk production	212
Reduced egg production	212
Poor eggshell quality	213
Infertility and poor fertility	214
Weakness and recumbency	215
Anemia and pallor	216
Bone abnormalities	216
Neurological signs	217
Diarrhea	218
Sudden death	219
Poor hatchability	219
Poor fish growth	220
Chapter 17 • Case Studies in Nutritional Diagnosis	222
How to work a nutritional case	224
What the seven cases share	232
Appendix A - Nutrient Deficiency Signs at a Glance	233
Appendix B - Nutrient Toxicity Signs	236
Appendix C - Mineral Antagonisms	237
Appendix D - Vitamin Interactions	239
Appendix E - Feed Quality Checklist	240
Appendix F - Nutritional Disease Investigation Form	241
Appendix G - Farm Nutritional Audit Checklist	242
Appendix H - Laboratory Investigation Checklist	243
Appendix I - Quick Reference: Abbreviations	244
Appendix J - Conversion Tables and Units	245
References and Further Reading	247
Glossary	248
The Author	250